Recovering navigable building graphs from IFC

Consolidated results record and measurement audit

BIM-Graphs / RoboBIM

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Abstract

This document consolidates every quantitative result the project currently stands behind, and records an audit of the measurements themselves. Two findings dominate. First, the project's long-standing headline — connectivity scored against human annotation — is *not* a correctness result: its specificity is exactly zero, because the annotation protocol can only record a negative where the pipeline already proposed a link, so a true negative is unrecordable. Second, two references that *can* discriminate were identified and scored: a 150-building generated corpus carrying 1,211 explicit negatives, on which the method beats its strongest baseline at $p < 10^{-6}$ at every damage level, and an annotation-free flood-fill reference over the real corpus which yields a real-corpus specificity of 0.644 — a number the project had previously concluded was unobtainable, on the strength of an argument that a later rebuild had already invalidated. The audit also corrects eight figures that had gone stale in the repository's own documentation, including the previous connectivity headline.

1 Scope and provenance

All figures below were produced on 18 September 2026 from the repository state of that date, using `/mnt/data/home/anedung/miniforge3/envs/bim-sg/bin/python` (Python 3.11.16, torch 2.5.1+cu121, transformers 5.16.1). Model inference ran on a single NVIDIA L40S (46 GB). Regression suites at time of writing: `tests/test_m0_regressions.py` reports 23/23 repairs holding and 4/4 open geometry findings still reproducing as intended; `tests/test_labels.py` and `tests/test_blind_render.py` pass.

Three classes of figure are distinguished throughout, because conflating them is how this project previously published numbers it had to retract:

- **Verified** — re-measured in this audit, command recorded.
- **Bounded** — cannot be computed exactly from the run that produced it; an interval is given instead of a point estimate.
- **Not quotable** — structurally incapable of measuring what its name suggests. Listed in §11 with the reason.

2 Corpora

All seven corpora pass `main/verify_corpus.py`, which checks each file is a structurally complete STEP file and matches a recorded size/SHA-256 manifest. This check exists because one corpus file was previously a truncated copy (12,845,056 bytes, exactly 3136×4096, ending mid-number with no `END-ISO-10303-21;`, missing 8 of its 19 doors) which IfcOpenShell opened without complaint, reporting a plausible 6 storeys and 35 spaces. It was annotated by a human before anyone noticed.

Corpus	Contents	Integrity
`dataset/test_set`	25 real IFC (the evaluation corpus)	pass
`dataset/2025_BIMfundamentals`	208 real IFC	pass
`ifc-graphx/annotator/data`	25 real IFC (deployed annotator)	pass
`synth150`	150 generated IFC, pristine	pass
`synth150_L2`	150, `IfcSpace` stripped from ∼half of storeys	pass
`synth150_L3`	150, spaces orphaned from their storey	pass
`synth150_L4`	150, no spatial containment, undeclared cm units	pass

Table 1: Source corpora, all manifest-verified.

Generated corpus. 150 buildings, 2,049 rooms, **1,708 positive pairs and 1,211 explicit adjacent-but-not-connected negatives**. The negatives are the design point: `main/generate_building.py` exposes a `door_ratio` precisely so that some adjacent room pairs are *not* joined, because if every neighbouring pair has a door then naive adjacency scores perfectly and the benchmark cannot separate any two methods.
Real corpus. 25 models yielding 1,027 rooms (681 stated as `IfcSpace`, 346 recovered geometrically). Of 146 storeys, only 74 contain any `IfcSpace`; `IfcRelSpaceBoundary` — the schema's own connectivity relation — occurs **zero** times across all 25 files, which is what makes this a recovery problem rather than a translation problem.

3 Connectivity against generated ground truth

This is the only corpus with enough explicit negatives to discriminate between methods. Ground truth is verified independently of the scoring pipeline by a raster flood-fill (`main/verify_gt.py`): **2,769 / 2,769 pairs agree**. That check rasterises one storey at a time, so those 2,769 are the same-storey decided pairs; the corpus decides 2,919 pairs in total and the remaining **150 are the cross-storey stair links, one per building, which this mechanism does not check**. Vertical connectivity in the generated reference therefore rests on the generator's own assertion.

	F_1 by damage level			
Method	L1	L2	L3	L4
`schema`	0.000	0.000	0.000	0.000
`adjacency`	0.696	0.635	0.635	0.635
`door_near`	0.916	0.858	0.858	0.858
`ifc2graph`	0.738	0.491	0.000	0.000
`ours`	**0.998**	**0.899**	**0.899**	**0.899**

Table 2: Pooled F_1 over 150 buildings per level. Reproduced at full floating-point precision by an independent pipeline run (§10).

3.1 Significance

Paired Wilcoxon signed-rank with a 20,000-resample bootstrap confidence interval on the paired difference, $n = 150$ buildings. A t-test is not used: F_1 is bounded and non-normal.

Level	Method	Precision	Recall	F_1	**Specificity**
L1	`schema`	0.000	0.000	0.000	1.000†
L1	`adjacency`	0.563	0.912	0.696	0.000
L1	`door_near`	1.000	0.845	0.916	1.000
L1	`ifc2graph`	0.585	1.000	0.738	0.000
L1	`ours`	**0.995**	**1.000**	**0.998**	**0.993**
L2	`adjacency`	0.546	0.757	0.635	0.113
L2	`door_near`	0.985	0.761	0.858	0.983
L2	`ifc2graph`	0.562	0.436	0.491	0.520
L2	`ours`	**0.983**	**0.827**	**0.899**	**0.980**

Table 3: Full metrics, L1 and L2. †`schema` asserts no edges at all, so it cannot produce a false positive and its specificity is 1.000 by definition. **Specificity must never be reported without recall beside it.**

Level	`ours` mean F_1	Strongest baseline	Δ	95% CI	W/L/T
L1	0.9980	`door_near` 0.9131	+0.0849	[+0.0743, +0.0962]	131/0/19
L2	0.8907	`door_near` 0.8502	+0.0405	[+0.0319, +0.0497]	81/2/67
L3	0.8907	`door_near` 0.8502	+0.0405	[+0.0319, +0.0497]	81/2/67
L4	0.8912	`door_near` 0.8502	+0.0410	[+0.0324, +0.0502]	81/2/67

Table 4: All four levels: $p < 10^{-6}$. Note that `main/significance.py` could not execute at all until 18 September 2026 — it called `sys.path.insert` with no `import sys`, raising `NameError` on import, and no test covered it. Every p-value quoted before that date predates the fix.

3.2 Two qualifications on the damage axis

The axis has two regimes, not four. L2, L3 and L4 yield identical pooled metrics for `adjacency`, `door_near` and `ours`. The files genuinely differ — on one building `IfcRelAggregates` drops $5 \to 4 \to 3$ and `IfcRelContained` $2 \to 0$ — but the graphs built from L2 and L3 are byte-identical for all three methods. Orphaning spaces from their storey (L3) is absorbed entirely, because storey assignment is recovered geometrically rather than read from containment; the undeclared centimetre units at L4 are absorbed by scale detection. This is a robustness result, but **no four-level severity gradient should be claimed for connectivity**. What moves the number is the L1→L2 step: missing `IfcSpace`, where half the storeys lose their rooms and recovery must supply them.

`ifc2graph` **collapses by construction, not on the merits.** It has no room-recovery stage, so once L2 removes `IfcSpace` those rooms are invisible to it, and at L3/L4 without spatial containment it emits nothing. Claiming a win over a published external tool at L3/L4 would be hollow; its L1 figure (F_1 0.738, specificity 0.000) is the fair one to quote.

4 Connectivity against the real corpus

Two independent references exist over the real corpus. They disagree about almost everything that matters, and the reason is instructive.

4.1 Human annotation — not a correctness result

15 buildings reviewed, 10 with complete verdict coverage.

Four qualifications, none of them optional:

1. **Specificity is exactly zero: all 53 reference negatives are also edges the pipeline**

Buildings	10
Reference positives / negatives / unknown	387 / 53 / 22
True positives / false negatives	374 / 13
False positives on a reference negative	53
Precision on pairs a reviewer ruled on	0.876
Recall	0.966
F_1	0.919
Strict precision / F_1 (charging out-of-scope and unmatched endpoints)	0.811 / 0.882
Macro mean per-building F_1	0.937
Specificity	**0.000**
Edge-type accuracy on matched pairs	**366/374 = 0.979**

Table 5: Annotated reference, re-measured 18 September 2026.

asserts. This is a property of the protocol, not of the method. The reference's own metadata states its scope — *"links the pipeline proposed and a reviewer rejected"* — so a rejection *is*, by construction, one of the pipeline's own false positives, and a true negative cannot be recorded at all. No discriminative metric is computable from this reference.

2. **Recall is a proposal-acceptance rate.** The reviewer only ever saw links the pipeline proposed; the reference's positives are the pipeline's own candidate set plus a handful of hand-added links. Room recall is worse still: all 375 reference rooms came from the pipeline and **zero** from an annotator, so room recall is 1.000 by construction and cannot detect a missed room at any rate.

3. **A trivial baseline wins on precision.** Joining every pair the reviewer examined scores 0.918 against 0.876. With 53 negatives against 387 positives, this slice is not discriminative.

4. **83% of judged items were swept in** with "the rest of this floor is correct". That is ground truth, but it was not inspected item by item.

The edge-type figure deserves its own note. It read 380/380 until 18 September 2026 and now reads 366/374. Composition copies the *pipeline's own* relation into the reference, so the metric compared the pipeline to itself and could only ever read 1.000. Restoring one building's 8 missing doors changed the pipeline's relation for 8 pairs the frozen reference still records as `open_passage` — and all 8 confusions are exactly `open_passage`→`connected_by_door`. The metric therefore measures neither correctness before nor after: it measures *whether the pipeline has changed since the reference was frozen.*

One building accounts for the degradation. `model_2` contributes 11 of the 13 false negatives and the single unmatched reference room, scoring recall 0.780 where every other building scores 1.000 or 0.949. Its verdicts were collected against the truncated file, whose missing doors had pushed its connectivity onto the `open_passage` mechanism. Re-judging the delta — 3 rooms, 13 links, 1 floor link — restores it.

4.2 The annotation-free derived reference — discriminative

`main/derive_real_gt.py` recovers connectivity from geometry alone by rasterising walls and flood-filling free space, a different mechanism from the pipeline's oriented door probing, and *independent of what the pipeline proposed.* It therefore does not share the defect above.

This reference was wrongly written off. The repository carried a section titled "why these numbers must not be used", whose principal argument was that 428 door-backed pairs disagreed with the reference against only 300 that agreed — a physically placed `IfcDoor` being

Models passing the reference's own trust gate	17 of 25
Positives / negatives / unknown	425 / 289 / 0
Precision	0.789
Recall	0.904
F_1	0.842
Specificity	**0.644**
Strict precision	0.406
Predictions out of scope / on unmatched endpoints	80 / 378

Table 6: Derived reference. Cross-validated two independent ways: this path (`evaluation.scoring.score_connectivity` via `match_rooms`) and `main/eval_real_connectivity.py`'s uncontracted mode (tp 384, fp 103) agree to the digit.

strong evidence of a connection. That argument predates a rebuild of the reference. Recomputed against the reference now on disk: **362 agree, 17 disagree**, a 96% reduction that nobody had re-checked. The conclusion drawn from the old data had outlived the data.

The residual disagreement is open finding A01, inside the ground-truth producer. Where a door has no authored `IfcOpeningElement` void, the wall's XY projection seals the doorway, the flood-fill calls the pair separated, and a correct prediction is charged as a false positive. That is the same projection-instead-of-section defect listed as an open method finding. It follows that repairing A01 would improve the method *and* the only independent real-corpus reference in one change — which argues for prioritising it above A03–A05.

5 Where the error actually is

Decomposing the annotated reference's verdicts by relation type is the single most actionable result in this document.

Relation	Confirmed	Rejected	Precision
`connected_by_door`	240	0	**1.000**
`vertically_connected`	67	0	**1.000**
`open_passage`	67	53	0.558

Table 7: Every measured error in the system falls in one relation.

Where IFC authors the answer — a door product, a stair solid — the deterministic path is *exact*. All 53 errors are in the one inferred relation. Moreover `open_passage` has no geometric signal in either direction: the maximum inscribed-circle diameter of the free gap measures 0.10 m for confirmed passages against 0.30 m for rejected ones on one building, 0.21 against 0.21 on another, and 0.01 against 0.00 on a third. It is measuring registration slop between `IfcSpace` footprints and wall faces and calling it connectivity. Relatedly, the width test uses an axis-aligned bounding box, which admits rejected passages with a median width of 3.51 m and a maximum of 15.74 m — a 15.74 m "doorway".

This is why the method should be **hybrid** rather than learned end to end: nothing can improve on 1.000, and the learnable problem is confined to the relation where geometry is silent.

6 Semantic labelling

Leave-one-out over the labels IFC did state, 592 eligible rooms from 21 models. The room's own label, and every feature derived from it, is removed before features are read.

Backend	Correct	Accuracy	vs. majority	Status
Majority class (`office`)	205/592	0.346	—	floor
`rule` (hand-written priors)	298/592	0.503	+0.157	verified
`llm` GLM-4-9B-0414	**329/592**	**0.556**	**+0.209**	verified
`vlm` GLM-4.1V-9B-Thinking	299/592	0.505	+0.159	**unreliable**

Table 8: The LLM figure is the first post-repair measurement of this row and exceeds the rule baseline by +0.052; `office` recall alone moves 0.698 → 0.839.

6.1 Why the VLM figure must not be quoted

Of 592 rooms, **95 (16.0%) hit the 300-token generation cap before producing a parseable answer**. On truncation, the backend falls back to the rule backend's own hypothesis — so those 95 "VLM answers" are the rule backend's answers wearing the model's name. The effect is the same size as the model's entire margin over majority-class guessing.

The run completed before this was instrumented, so the exact split cannot be recovered from it. It can only be bounded. With 299 correct of 592, and 95 answers borrowed:

$$\frac{299-95}{592-95} = 0.410 \quad \leq \quad \text{accuracy}_{\text{model only}} \quad \leq \quad \frac{299}{592-95} = 0.602.$$

That interval spans from below the rule baseline to above the text LLM, so the ordering of the three backends is *not* established by this run. `main/eval_semantics.py` now records `truncated_correct` and `accuracy_excl_truncated` per run, so a re-run reports the honest figure directly. **It should be re-run before this row is used.**

6.2 Per-class recall

Class	n	`rule`	`llm`	`vlm`
`office`	205	0.698	**0.839**	0.693
`bathroom`	132	0.530	0.538	0.530
`circulation`	86	0.256	0.233	0.256
`stair`	45	0.644	0.644	0.622
`bedroom`	28	0.786	0.786	0.786
`meeting`	23	0.174	0.174	0.174
`living`	22	0.182	0.273	0.318
`kitchen`	18	0.222	0.222	0.222
`outdoor`	10	0.000	0.000	0.000
`storage`	10	0.000	0.100	0.000
`retail`	8	0.000	0.000	0.000
`service`	5	0.000	0.000	0.000

Table 9: Four of twelve classes never resolve, each with ≤ 10 held-out examples. Whether that is too little evidence or a real failure is not established. The dominant confusion in all three backends is `bathroom`→`storage` (52 cases).

Held-out rooms are stated `IfcSpaces`, which are geometrically cleaner than the recovered rooms this stage actually serves. All figures here are therefore an **optimistic bound**, and the 346 recovered rooms have no accuracy figure of their own.

7 Goal resolution

Two qualifications. First, **this is not an independent measurement of retrieval**: per-class recall here is identical to rule-backend label accuracy for 11 of 12 classes, because resolving "go

Resolution accuracy	298/592 = 0.503
Reachable, wheeled profile	91/298 = 0.305
Reachable, legged profile	252/298 = 0.846
Reachable, human profile	252/298 = 0.846

Table 10: Re-measured 18 September 2026. The previously circulated figure of 0.519 came from a file quarantined for a serialization defect and should not be cited.

to the kitchen" mostly succeeds or fails on whether the room was labelled correctly, not on the retrieval step. Second, the `legged` and `human` profiles report **byte-identical** counts, 252/298 each. That reproduces open finding A05 — feasibility is two-valued, so a route can rest on an unverified connection — in a fresh measurement rather than only in a regression fixture.

8 The vision modality

A leak-free image corpus was built so that a model given a picture is solving a spatial reasoning problem rather than an OCR problem.

Corpus	Condition	Images	Note
Real (208 models)	architectural plan	966	walls, doors, windows
Real (208 models)	occupancy raster	984	binary, 0.10 m/cell
Generated (4×150)	architectural plan	1,200	300 per level
Generated (4×150)	occupancy raster	1,200	300 per level

Table 11: Produced by `main/render_blind.py`, gated by `tests/test_blind_render.py`.

The pre-existing image corpus was unusable and every image had to be replaced. The previous renderer burned the answer into the pixels: room names and areas in legible type, a `?` appended to every geometrically recovered room, and a fill tint distinguishing IFC-stated from pipeline-recovered rooms — i.e. part of the ground-truth partition. Its `--no-labels` flag suppressed only the first two. One ground-floor plan contained 949,601 pixels of the recovered-room fill colour alone. The new renders contain at most 4 such pixels, which is antialiasing coincidence; the leak gate is validated by being run against the old images, where it fires.

Doors are now visually distinct from windows. The previous renderer placed `IfcDoor` and `IfcWindow` in one style bucket at identical colour and weight, making door-mediated connectivity unrecoverable from vision and the text-vs-vision comparison unfair — the text modality receives the door products explicitly.

The modality comparison is controlled, and this is measured. All 150 generated buildings have byte-identical drawable geometry across L1–L4, and all 600 rendered images (300 plan, 300 occupancy) are **byte-identical across all four damage levels**. `IfcWall` count is 8,912 at every level while `IfcSpace` drops 2,049 → 994. The text modality therefore loses half its rooms while the picture does not change by a single pixel, so any score difference across levels in a vision condition cannot be attributed to the image.

One stated limitation. The occupancy raster is not bounded to the storey's floor plate, because no `IfcSlab` survives a 1.2 m section cut. Free space therefore continues past the building envelope wherever a real opening exists, including through external doors. It is an image for a model to read, not a segmentation mask.

Outcome	Rooms	Share
Mapped to one of 12 classes	5,158	81.6%
Unnamed (no functional content: “Room”, “Raum”, “Area”)	459	7.3%
Unmapped (a real function outside the taxonomy: “Gym”, “Studio”)	583	9.2%
Ambiguous (combined function: “Living room / Kitchen”)	123	1.9%
Of the 5,864 rooms carrying a real name: **5,158 (88.0%) map**		

Table 12: 6,323 IFC-stated rooms across the 208-model corpus, `evaluation/labels.py` v1.0.0.

9 The label vocabulary

The previous implementation answered a three-way question with two answers, returning “no class” both for a string carrying no function and for a real function the taxonomy could not express — and every caller discarded both identically, so the taxonomy’s blind spots were invisible. The sharpest consequence: the generator emits `Lab` as one of twelve synthetic room types and no keyword matched it, so **166 of 2,049 rooms in the primary corpus (8.1%) were silently unscorable**. That is now zero, and a regression test fails if the generator gains a room type the vocabulary does not know.

Two resolution defects were also fixed: punctuation was never stripped, so “W.C. (Women)” did not match the key `wc`; and keyword matching took the first hit in dictionary order, so “Store Front” resolved to `storage` rather than `retail`. Longest-match was tried as a remedy and *rejected* — it sends “Lift Shaft” to `service` rather than `stair`, and a lift is vertical circulation. The ten genuine conflicts are resolved by an explicit table instead.

10 Reproducibility and determinism

Running the pipeline’s own single entry point a second time (`./run_pipeline.sh --regenerate 150`) reproduced all 20 (method × level) connectivity cells **identically at full floating-point precision**, not merely to three decimals.

What that does and does not show. Checked against the archived corpus: the 150 ground-truth files are **byte-identical 150/150**, while the 150 IFC files are **byte-identical 0/150** — every input file differs in its GUIDs and header, and the ground truth it encodes does not. So this demonstrates that the parse → recover → score chain is invariant to IFC identifier churn, which is worth having. It is **not** a fresh sample of buildings and says nothing about variance across different layouts: the same seed yields the same rooms by construction. A sampling-variance claim would require varying the generator seed, which has not been run.

11 Metrics that are not quotable

Recorded explicitly, because each reads like a result and is not one.

Metric	Why not
Specificity on the annotated reference	Structurally 0.000. A negative exists only where the pipeline proposed a link and a reviewer rejected it, so a true negative is unrecordable.
Recall on the annotated reference	A proposal-acceptance rate: the reference’s positives are the pipeline’s own candidate set plus a few hand-added links.
Room recall on the annotated reference	1.000 by construction. All 375 reference rooms came from the pipeline, none from an annotator.

Metric	Why not
Edge-type accuracy	Composition copies the pipeline's own relation into the reference. It measures whether the pipeline changed since the reference was frozen.
`schema` specificity (1.000)	It asserts no edges, so it cannot produce a false positive.
`ifc2graph` at L3/L4 (0.000)	No room-recovery stage; the rooms are invisible to it by construction, not on the merits.
VLM label accuracy (0.505)	16.0% of answers are the rule backend's, borrowed on truncation. True value bounded to $[0.410, 0.602]$.
Intra-storey reachability	Rewards over-connection: the `adjacency` baseline, which joins through solid walls, scores 0.935 against 0.745.
Goal resolution as retrieval	Per-class recall is identical to label accuracy for 11 of 12 classes.
Any figure from the pre-repair snapshot	Quarantined for a serialization defect; the directory's own README is headed "do not quote these".

12 Open findings and necessary conditions

Four geometry findings remain unrepaired, and the regression suite asserts they still reproduce, so a fix fails its case and must be flipped deliberately.

ID	Status	Description
A01	open	Room recovery takes the XY projection, not a section, so a lintel seals a doorway. Also present *inside* the derived reference, where it is the cause of the residual door-backed disagreement.
A03	open	Door probing samples at fixed heights without checking the intervening segment, and reaches past a 0.5 m room through a wall.
A04	open	Reported width is the longest world-axis bounding-box side: a 1.0 m door rotated 45° reports 0.742 m. The call site that matters admits a 15.74 m passage.
A05	open	Feasibility is two-valued; a missing width reads as passable, and `legged` and `human` reachability are byte-identical.

Necessary-condition checks over the 25 real models (`main/validate.py`) — these can falsify a graph but cannot rank one: `R2` doors-join-two-rooms 25/25, `R3` one-storey-per-room 25/25, `R4` plausible dimensions 25/25, `R6` containment sanity 25/25; `R1a` horizontal 9/25, `R1b` vertical 10/24 (1 n/a), `R5` coverage 18/25. Mean intra-storey 0.745, mean reachability 0.742, mean element coverage 0.634.

Two contract gaps remain open. There is no split manifest, so a threshold tuned on a building can still be evaluated on it — and four of the ten annotated buildings are provably development material, cited by name in source comments. And there is no schema validator run on every output: a graph declaring a nonexistent schema version currently scores F_1 1.000, and an empty graph records as complete with exit status zero.

13 Corrections made in this audit

Eight figures in the repository's own documentation were stale or wrong. All are now corrected at source, with the superseded values retained where they form a dated record.

Claim	Was	Now, and why
Annotated-reference headline	0.878 / 0.982 / 0.927, strict 0.903	0.876 / 0.966 / 0.919, strict 0.882. One building rebuilt from its intact source IFC.
Edge-type accuracy	380/380	366/374. The 8 confusions are all `open_passage`→`connected_by_door`, from that same rebuild.
Per-relation decomposition	240 doors read as 230; passage 83/53 = 0.610	door 240/0, vertical 67/0, passage 67/53 = 0.558. The qualitative finding strengthened: restored doors moved connections off the failing mechanism.
LLM margin over rule	+0.053	+0.052. Arithmetic error (31/592 = 0.0524).
Regression suite size	"22 repairs"	23. Documentation drift in two files.
Reproducibility framing	"a completely fresh corpus"	Ground truth is byte-identical 150/150; only the IFC serialization differs. Demonstrates determinism, not sampling variance.
Ground-truth verification scope	"2,769/2,769 pairs"	Same-storey pairs only. The 150 cross-storey stair links are not checked by that mechanism.
Derived reference usability	"these numbers must not be used" (428 vs 300 door-backed disagreements)	17 vs 362 after a rebuild the argument predates. The reference is discriminative and now carries the project's only real-corpus specificity.

A ninth issue is recorded but not fixed: the VLM result record in `results/semantics_eval.json` predates the truncation instrumentation, so it reports 0.505 with no truncation field. A reader consulting only the JSON would see no warning.

14 What the evidence currently supports

1. **On a reference that can discriminate, the method wins decisively.** F_1 0.998 and specificity 0.993 at L1, beating the strongest baseline by +0.085 with $p < 10^{-6}$ and 131 wins to 0 losses. Under the damage that matters — missing `IfcSpace` — it degrades to 0.899 and still wins at $p < 10^{-6}$.

2. **The real-corpus evaluation everyone had been quoting was the problem, not the method.** On the annotated reference the same contrast gives $p = 0.805$ with the ablation ahead; on a reference with genuine negatives the ordering reverses. Specificity 0.000 versus 0.644 is the whole story.

3. **Error is confined to one mechanism.** Door- and stair-mediated connectivity are exact (1.000, 1.000). All 53 measured errors are `open_passage`, where the geometric signal is absent in both directions. This localises the research problem precisely and argues for a hybrid method rather than a learned one.

4. **Repairing A01 pays twice** — it is a defect in the method and in the only independent real-corpus reference simultaneously, which should lift it above A03–A05 in priority.

5. **Of the learned results, exactly one is currently trustworthy**: the text LLM at 0.556 against a 0.346 floor and a 0.503 rule baseline. The vision result is not usable until re-run.

Reproduce the connectivity results with `./run_pipeline.sh --regenerate 150`; the semantic results with `python main/eval_semantics.py --backend {rule,llm,vlm}`; and verify corpus integrity first with `python main/verify_corpus.py <dir>`. Regression suites: `tests/test_m0_regressions.py`, `tests/test_labels.`
`tests/test_blind_render.py`.